Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Harrison Cottom	hcottom@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

I am a civil engineer and I want to create an application that can determine the best traffic light timing for different intersections based on traffic demand (vehicles per hour). In the future I want to be a traffic analyst or highway engineer.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are many websites and software that can be used to simulate traffic flow through intersections. Some like PTV Vistro and Synchro that help engineers analyze intersections to develop signal timing plans use very advanced features that I will not be able to recreate. I will try to instead focus on using traffic volumes and intersection patterns to calculate the best green-light timing.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I would like this website to be used to help optimize the best traffic signal time for different

traffic demands to allow for a better understanding of an intersection's estimated light timings without needing to conduct long-term observation. My site would not be used as a replacement to traditional tools, but as a way of focusing specifically on the traffic light timing possibly for education purposes. By allowing users to save multiple different scenarios for comparison the site would be useful for anyone trying to better understand traffic light optimization.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program should take traffic volume data from multiple directions at an intersection using those values to calculate the optimal available green time between traffic signals, finally displaying the optimal time and saving the scenario.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

Users will need to select an intersection name, the associated VPH in all directions of the intersection, and the total length of the signal cycle in seconds.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will filter out any inputs with negative signal cycle or VPH values as that is not a possible scenario. To get the actual output the computer should add the two traffic volumes together then divide each input by the total volume to determine what percentage of the signal each intersection will require. All inputs and calculated percentage and optimal traffic light timing would be saved as a dataclass so the user can look back on it later.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The site would display the given traffic volumes, the calculated traffic percentage for each direction, and the suggested green light timing in seconds. Users will be able to save scenarios so they can go back and compare their results if the intersection has a variable traffic rate. A disclaimer above the suggestion will note that values do not account for many other factors and should not be used as a replacement for traditional tools. Along with this previously calculated scenarios will have bar graphs created and temporarily saved to a section for comparison viewing.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

My program would track traffic volume, intersection name, and intersection patterns.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details I need: Intersection name, traffic volume in all directions in vehicles per hour, and intersection pattern

Details I will Ignore: Exact vehicle positions and driver behaviour, road geometry and lane configuration, weather and unexpected road accidents, pedestrians and other miscellaneous signal phases.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

I think that the most complex parts of this project will revolve around meeting the project criteria of 5 pages minimum so I will have to get creative with potentially making a page for

saved calculations and pages for each different type of intersection. Another aspect that will be challenging is figuring out how to save the scenarios as dataclasses and displaying them on the saved page. To ensure I have two dataclasses I could use nested dataclasses within the scenario dataclass so that it can fit all different types of intersections.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

To help avoid that I understand how the code works for each section that I use the LLM for. I will ask it to make me a template for what the code should look like so I can first ask questions about the format before customizing it to my project. With the help of the provided trained LLM I can also have it ask me questions before giving me the answer so that I can build or prove my understanding of the code.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal for this project is to design a website that I can mess around with in my free time when I am thinking about different road designs in my head. I often find myself wondering why certain signals take so long and this could give me a rough understanding of why certain lights are timed in specific ways.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

use the above mentioned project to respond to this prompt as if you were an engineering freshman "I want to build a simple web tool to help students track their study hours. What

are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The LLM mentioned that traffic lights often give too much green time to a direction with fewer cars, traffic volumes are often different throughout a day, and it can be challenging to compare different saved estimations without extensive calculation. It suggested that the website should have an input for each direction's traffic volume along with a calculator that divides the available green time based on each direction's percentage of traffic, and finally, a feature to save and compare different scenarios. Two of the 3 mentioned features were already established prior to the prompt. But the idea about using a potential existing green timing with the calculated percentages does sound good.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

I want to do a green time comparison chart for the saved scenarios but that might be too complex for CS1 do you believe there is a simple way to do it

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

I think that I want to focus more on the comparison aspect with a bargraph feature that allows visualization of previous scenarios so that comparison wouldn't be as complex. This would also add another page to my program while not requiring length coding.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck I will utilize the internet and previous textbook modules to see if I can figure out a solution on my own before consulting the LLM to see if it can guide me to the right solution without directly giving me the answer.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I will first read the error message to see if I can decipher it without the help of the LLM by stepping through the code near the error to see if I can troubleshoot it on my own. If I cannot find the source of the error on my own, I will then ask the LLM what the error message could mean and check for what it responds with. If the code still will not run I will make a post on the CS1 discord or attend TA office hours to receive help.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I cannot copy and paste the whole conversation as it is pretty long, but I can provide a link with my assignment that leads to the conversation.
https://chatgpt.com/share/6ae3aa5f-e498-83ea-9fda-0cd206a88b03

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.